X-ray constrained (or restrained) wavefunctions

Dylan Jayatilaka

Technical University of Denmark (DTU)

University of Western Australia (UWA)

1 June 2025

X-ray Constrained Wavefunctions

1. What is a wavefunction?
2. Wavefunctions and nuclei
3. Electron coordinates and spin
4. Calculating the wavefunction
5. The simplest wavefunction: Hartree-Fock
6. The Hohenberg-Kohn theorem
7. Experimentally fitted wavefunctions
8. Some examples

X-ray Constrained Wavefunctions

X-ray Constrained Wavefunctions

1. What is a wavefunction?
2. ~~Wavefunctions and nuclei~~
3. ~~Electron coordinates and spin~~
4. Calculating the wavefunction
5. The simplest wavefunction: Hartree-Fock
6. The Hohenberg-Kohn theorem
7. Experimentally constrained wavefunctions
8. Some examples

X-ray Constrained Wavefunctions

Slides and updated lecture notes are on the `tonto-chem` repository at `github`, the `release-td` repository.

Or you can go here directly:

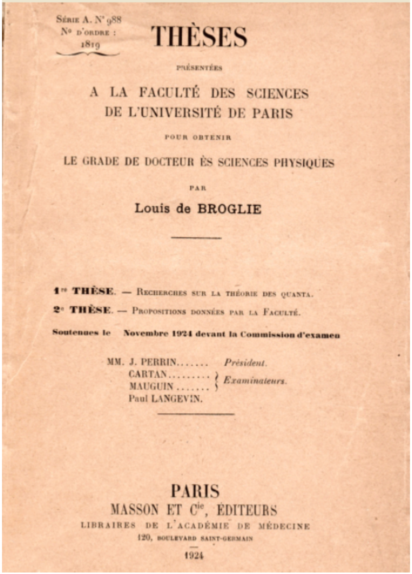
`github.com/dylan-jayatilaka/tonto/tree/release-td`

This year is 100 years of quantum mechanics!

- Louis de Broglie's thesis was published in 1925 in *Ann. de Physique* outlining his idea of matter waves.

  Strictly speaking, he passed in 1924.

- Do you notice who were the examiners?



SÉRIE A. N° 988
N° D'ORDRE : 1819

THÈSES

PRÉSENTÉES

A LA FACULTÉ DES SCIENCES
DE L'UNIVERSITÉ DE PARIS

POUR OBTENIR

LE GRADE DE DOCTEUR ÈS SCIENCES PHYSIQUES

PAR

Louis de BROGLIE

1re THÈSE. — Recherches sur la théorie des quanta.
2e THÈSE. — Propositions données par la Faculté.

Soutenues le   Novembre 1924 devant la Commission d'examen

MM. J. PERRIN....... *Président.*
CARTAN.......... } *Examinateurs.*
MAUGUIN........ }
Paul LANGEVIN.

PARIS
MASSON ET Cie, ÉDITEURS
LIBRAIRES DE L'ACADÉMIE DE MÉDECINE
120, BOULEVARD SAINT-GERMAIN
1924

1. What is the wavefunction?

The *wavefunction* describes the state of a system in quantum mechanics.

The wavefunction holds all the information needed to describe an imagined system in detail.

1. What is the wavefunction?

How it is written, in more detail

$$\begin{aligned}\Psi &\equiv \Psi(\boldsymbol{x}; \boldsymbol{R}; t) \\ &\equiv \Psi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{N_\mathrm{e}}; \boldsymbol{R}_1, \ldots, \boldsymbol{R}_{N_\mathrm{N}}; t)\end{aligned}$$

$$\begin{aligned}\boldsymbol{x} &= \text{are the electron coordinates} \\ \boldsymbol{R} &= \text{are the nuclear coordinates} \\ t &= \text{is the single time coordinates}\end{aligned}$$

1. What is the wavefunction?

How it is written, in more detail

$$\begin{aligned}\Psi &\equiv \Psi(\boldsymbol{x}; \boldsymbol{R}; t) \\ &\equiv \Psi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{N_\mathrm{e}}; \boldsymbol{R}_1, \ldots, \boldsymbol{R}_{N_\mathrm{N}}; t)\end{aligned}$$

$\boldsymbol{x}$ = are the electron coordinates

$\boldsymbol{R}$ = are the nuclear coordinates

t = is the single time coordinates

The nuclei are normally assumed to be classical point-like particles in the first instance, which is called the *Born-Oppenheimer (BO) approximation*.

Here, Ψ is a BO wavefunction.

This means nuclear coordinates $\boldsymbol{R}$ in Ψ are parameters i.e. the “wave” motion in the BO wavefunction only applies to the electron coordinates $\boldsymbol{x}$.

Chemistry is mostly based on the idea that nuclei are classical point particles — *but they are not*.

1. What is the wavefunction?

How it is written, in more detail

$$
\begin{aligned}
\Psi &\equiv \Psi(\boldsymbol{x}; \boldsymbol{R}; t) \\
&\equiv \Psi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{N_\mathrm{e}}; \boldsymbol{R}_1, \ldots, \boldsymbol{R}_{N_\mathrm{N}}; t)
\end{aligned}
$$

$\boldsymbol{x} =$ are the electron coordinates

$\boldsymbol{R} =$ are the nuclear coordinates

$t =$ is the single time coordinates

Ψ is generally a *complex-number valued function* of the particle coordinates in the system.

* Though for non-generate time-independent ground states, which is the most common case, Ψ can be chosen to be a real function

1. What is the wavefunction?

How it is written, in more detail

$$\Psi \equiv \Psi(\boldsymbol{x}; \boldsymbol{R}; t)$$
$$\equiv \Psi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{N_\mathrm{e}}; \boldsymbol{R}_1, \ldots, \boldsymbol{R}_{N_\mathrm{N}}; t)$$

$\boldsymbol{x}$ = are the electron coordinates

$\boldsymbol{R}$ = are the nuclear coordinates

t = is the single time coordinates

Born rule: the probability of finding the first electron at coordinates $\boldsymbol{x}_1$ is given by:

$$p_\mathrm{e}(\boldsymbol{x}_1; \boldsymbol{R}) = \int |\Psi(\boldsymbol{x}; \boldsymbol{R})|^2 d\boldsymbol{x}_2..d\boldsymbol{x}_{N_\mathrm{e}},$$

$$\int p_\mathrm{e}(\boldsymbol{x}_1; \boldsymbol{R}) d\boldsymbol{x}_1 = 1.$$

Normalisation rule for any probability density

1. What is the wavefunction?

How it is written, in more detail

$$\Psi \equiv \Psi(\boldsymbol{x}; \boldsymbol{R}; t)$$
$$\equiv \Psi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{N_\mathrm{e}}; \boldsymbol{R}_1, \ldots, \boldsymbol{R}_{N_\mathrm{N}}; t)$$

$\boldsymbol{x}$ = are the electron coordinates

$\boldsymbol{R}$ = are the nuclear coordinates

t = is the single time coordinates

But electrons are indistinguishable! So we must use the *density of all electrons in the system*,

$$\rho_\mathrm{e}(\boldsymbol{x}; \boldsymbol{R}) = N_\mathrm{e}\, p_\mathrm{e}(\boldsymbol{x}; \boldsymbol{R}),$$
$$\int \rho_\mathrm{e}(\boldsymbol{x}; \boldsymbol{R}) d\boldsymbol{x} = N_\mathrm{e}.$$

1. What is the wavefunction?

How it is written, in more detail

$$\Psi \equiv \Psi(\boldsymbol{x}; \boldsymbol{R}; t)$$
$$\equiv \Psi(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{N_\mathrm{e}}; \boldsymbol{R}_1, \ldots, \boldsymbol{R}_{N_\mathrm{N}}; t)$$

$\boldsymbol{x}$ = are the electron coordinates

$\boldsymbol{R}$ = are the nuclear coordinates

t = is the single time coordinates

Questions:

- Why did I write the electron coordinates $\boldsymbol{x}$'s and not $\boldsymbol{r}$ like in classical physics?
- Is it correct to write only one time coordinate t when all the particles have different positions?

  Why not also different times?

2. Wavefunctions and nuclei?

2. Wavefunctions and nuclei?

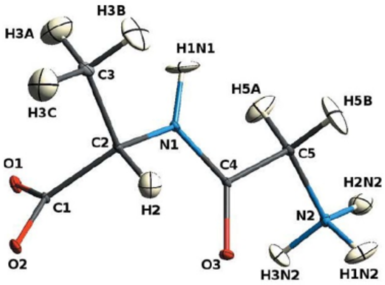


Figure 1: Structure of Gly-L-Ala determined by X-ray diffraction at 12K showing the probability distributions associated with the nuclear positions. Even in a perfect crystal at low temperature these are present and indicate that, lke the electrons, the nuclei should also be described by (phonon) wavefunctions or density matrices. From Capelli et al (2014).

You have probably seen pictures of molecules in molecular crystal structures?

Then you would have noticed that the nuclei in the molecules appear like **Australian footballs** — *but why*?

2. Wavefunctions and nuclei?

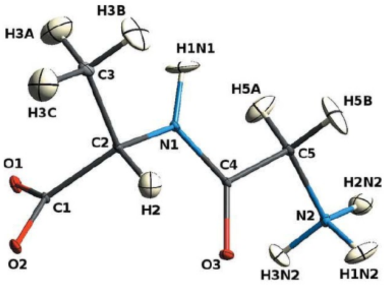


Figure 1: Structure of Gly-L-Ala determined by X-ray diffraction at 12K showing the probability distributions associated with the nuclear positions. Even in a perfect crystal at low temperature these are present and indicate that, lke the electrons, the nuclei should also be described by (phonon) wavefunctions or density matrices. From Capelli et al (2014).

You have probably seen pictures of molecules in molecular crystal structures?

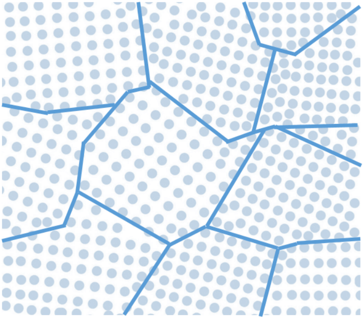
Then you would have noticed that the nuclei in the molecules appear like **Australian footballs** — *but why*?

The reason is

— that the nuclei are wavelike particles and are delocalised, we should model their probability distributions.

— the nuclei in different unit cells are not quite in translationally equivalent sites, *the crystallites are misaligned*.

2. Wavefunctions and nuclei?



Here is an extreme example of misaligned crystallites in a metallic crystal.

In fact, to have good Bragg data (i.e. to avoid “extinction” or “dynamical scattering”) *we need perfectly imperfect crystals*.

— If not, we won’t have a good chance to correctly extract the QM part of the nuclear vibration football.

2. Wavefunctions and nuclei?

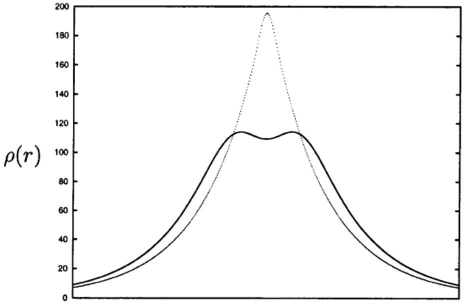


Fig. 3. The electron density along the bond axis in $X\ {}^1\Sigma_g^+\ N_2$ molecule, showing a 0.5 atomic units slice around one N atom for the static charge density ρ_e (*dotted line*), and for the vibrationally averaged charge density, $\bar{\rho}$, in the $v = 1$ state (*solid line*)

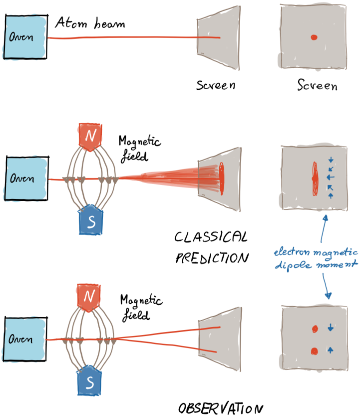
From Cassam-Chenai & Jayatilaka (2001) TCA 105 p. 213

Here is some fun.

- Why does the vibrationally averaged electron density on the N atom in the N_2 molecule in its first vibrational state have a *double peak?*

3. Electron coordinates $\boldsymbol{x}$ and spin

3. Electron coordinates $\boldsymbol{x}$ and spin



The reason we used $\boldsymbol{x}$ for the electron coordinates is that it hides another coordinate:

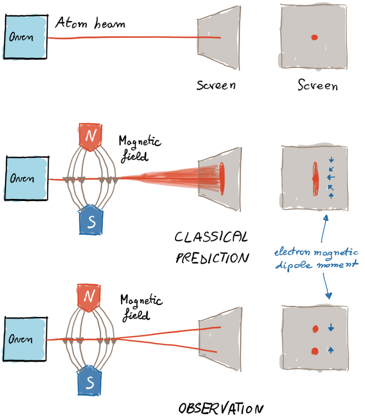
$$\boldsymbol{x} = (\boldsymbol{r}, \xi)$$

This is the *spin coordinate* ξ.

— ξ is continuous but the wavefunction associated with it is *two dimensional*; these two components are called "up" (or α) and "down" (or β).

— By comparison $\boldsymbol{r}$ is continuous but the wavefunction associated with it is only one dimensional (scalar valued).

3. Electron coordinates $\boldsymbol{x}$ and spin



The reason we used $\boldsymbol{x}$ for the electron coordinates is that it hides another coordinate:

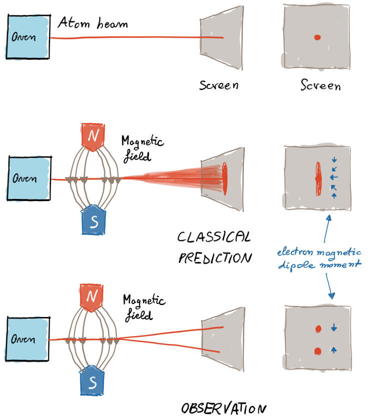
$$\boldsymbol{x} = (\boldsymbol{r}, \xi)$$

This is the *spin coordinate* ξ.

— The notion that electrons have this extra dimension was suggested by the famous Stern-Gerlach experiment.

— Pauli also explained theoretically how this extra dimension appears quantum theory, and that it must be associated with an intrinsic magnetic moment, which may take on only two opposite values, as is observed.

3. Electron coordinates $\boldsymbol{x}$ and spin



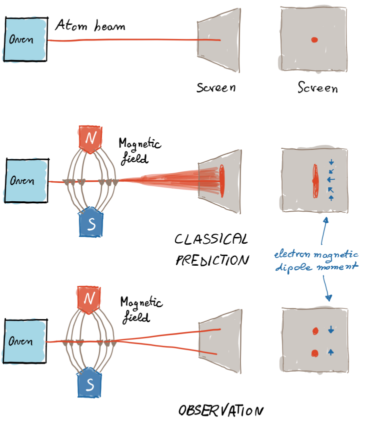
The reason we used $\boldsymbol{x}$ for the electron coordinates is that it hides another coordinate:

$$\boldsymbol{x} = (\boldsymbol{r}, \xi)$$

This is the *spin coordinate* ξ.

— More support for the existence of the two component nature spin is that it allows to explain atomic structure – that two electrons may occupy the same region of space if they have different spin components – and this then explains the whole periodic table.

3. Electron coordinates $\boldsymbol{x}$ and spin



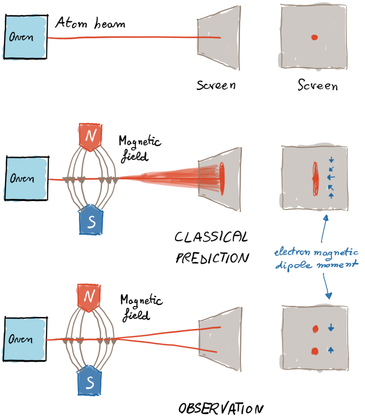
The reason we used $\boldsymbol{x}$ for the electron coordinates is that it hides another coordinate:

$$\boldsymbol{x} = (\boldsymbol{r}, \xi)$$

This is the *spin coordinate* ξ.

— The two component nature of the spin coordinate permits to explain the ground state electronic structure of all atomis – by allowing two electrons to occupy the same region of space, provided they have opposite spin components.

3. Electron coordinates $\boldsymbol{x}$ and spin



The reason we used $\boldsymbol{x}$ for the electron coordinates is that it hides another coordinate:

$$\boldsymbol{x} = (\boldsymbol{r}, \xi)$$

This is the *spin coordinate* ξ.

— And the clincher: with the spin coordinate, extremely accurate QM calculations can be done e.g. those by Kołos & Wolniewicz, which are some of the most accurate and precise predictions from QM ever made.

3. Electron coordinates *x* and spin

Table 3. Dissociation Energies (in cm^{-1}) of H_2 and D_2 in Their Ground State—A Comparison with the Experiment

	D_0/cm^{-1}	
	H_2	D_2
experiment (1993)	36118.06(4)[a]	36748.32(7)[a]
experiment (2004)	36118.062(10)[b]	36748.343(10)[b]
experiment (2009/10)	36118.06962(37)[c]	36748.36286(68)[d]
theory	36118.0696(11)	36748.3634(9)
difference	0.0000(12)	0.0005(11)

[a] Ref 66. [b] Ref 67. [c] Ref 1. [d] Ref 2.

Table 4. The Lowest Rotational and Vibrational Excitation Energies (in cm^{-1})

	H_2	
	$J = 0 \to 1$	$v = 0 \to 1$
theory	118.486812(9)	4161.1661(9)
experiment	118.48684(10)[a]	4161.1660(3)[b]
difference	−0.00003(10)	0.0001(9)
	D_2	
	$J = 0 \to 1$	$v = 0 \to 1$
theory	59.780615(3)	2 993.6171(2)
experiment	59.78130(95)[c]	2 993.6130(19)[d]
difference	−0.00068(95)	0.0041(19)

[a] Ref 68. [b] Ref 46. [c] Ref 2. [d] Ref 45.

Komasa et al (2011) JCTC 7 p. 3105

This is to show you how good some of the predictions can be.

There are about hundred other results like this.

They are rarely mentioned by physicists—why not?

Anyway: one starts to believe that this QM is a rather good . . . for chemistry . . .

3. Electron coordinates $\boldsymbol{x}$ and spin

Pauli's 1927 paper & Dirac's 1926 paper

3. Electron coordinates *x* and spin

Pauli's 1927 paper & Dirac's 1926 paper

" Zur Quantenmechanik des magnetischen Elektrons," Zeit. f. Phys. **43** (1927), 601-623.

On the quantum mechanics of magnetic electrons

By **W. PAULI, Jr.** in Hamburg

(Received on 3 May 1927)

Translated by D. H. Delphenich

moment in a fixed direction as further independent variables in order to carry out the computations of its rotational degrees freedom, along with the position coordinates of any electron. In contradiction to classical mechanics, these variables can assume only the variables $+\frac{1}{2}\frac{h}{2\pi}$ and $-\frac{1}{2}\frac{h}{2\pi}$, which is completely independent of any sort of external field. The appearance of the aforementioned new variables thus implies a simple splitting of the eigenfunctions into two position functions ψ_α, ψ_β for one electron, and more generally, for N electrons they split into 2^N functions, which are to be regarded as the "probability amplitudes" that in a well-defined stationary state of the system not only do the position coordinates of the electrons lie in a given infinitesimal interval, but also that the components of their proper moments in the chosen direction should have the given values, which are $+\frac{1}{2}\frac{h}{2\pi}$ for ψ_α and $-\frac{1}{2}\frac{h}{2\pi}$ for ψ_β. Methods

$$\left.\begin{array}{l}\psi^{\text{sym.}}\alpha_1\alpha_2(q_1,q_2)=\psi^{\text{sym.}}\alpha_1\alpha_2(q_2,q_1),\ \psi^{\text{sym.}}\alpha_1\beta_2(q_1,q_2)=\psi^{\text{sym.}}\beta_1\alpha_2(q_2,q_1),\\ \psi^{\text{sym.}}\beta_1\alpha_2(q_1,q_2)=\psi^{\text{sym.}}\beta_1\alpha_2(q_2,q_1),\ \psi^{\text{sym.}}\beta_1\beta_2(q_1,q_2)=\psi^{\text{sym.}}\beta_1\beta_2(q_2,q_1),\end{array}\right\} \tag{26'}$$

$$\left.\begin{array}{l}\psi^{\text{antis.}}\alpha_1\alpha_2(q_1,q_2)=-\psi^{\text{antis.}}\alpha_1\alpha_2(q_2,q_1),\ \psi^{\text{antis.}}\alpha_1\beta_2(q_1,q_2)=-\psi^{\text{antis.}}\beta_1\alpha_2(q_2,q_1),\\ \psi^{\text{antis.}}\beta_1\alpha_2(q_1,q_2)=-\psi^{\text{antis.}}\beta_1\alpha_2(q_2,q_1),\ \psi^{\text{antis.}}\beta_1\beta_2(q_1,q_2)=-\psi^{\text{antis.}}\beta_1\beta_2(q_2,q_1).\end{array}\right\} \tag{27'}$$

The skew-symmetric solution is also the one that fulfills the "equivalence rule" in the general case of N electrons, and is the only one that occurs in nature [2]). It seems to me to

Here is Pauli's 1927 paper which suggested that electrons have an two component form, associated with angular momentum.

He was the one who named them "α" and "β".

The Nobel-prize winning sentence is underlined.

3. Electron coordinates *x* and spin

Pauli's 1927 paper & Dirac's 1926 paper

" Zur Quantenmechanik des magnetischen Elektrons," Zeit. f. Phys. **43** (1927), 601-623.

On the quantum mechanics of magnetic electrons

By **W. PAULI, Jr.** in Hamburg

(Received on 3 May 1927)

Translated by D. H. Delphenich

moment in a fixed direction as further independent variables in order to carry out the computations of its rotational degrees freedom, along with the position coordinates of any electron. In contradiction to classical mechanics, these variables can assume only the variables $+\frac{1}{2}\frac{h}{2\pi}$ and $-\frac{1}{2}\frac{h}{2\pi}$, which is completely independent of any sort of external field. The appearance of the aforementioned new variables thus implies a simple splitting of the eigenfunctions into two position functions ψ_α, ψ_β for one electron, and more generally, for N electrons they split into 2^N functions, which are to be regarded as the "probability amplitudes" that in a well-defined stationary state of the system not only do the position coordinates of the electrons lie in a given infinitesimal interval, but also that the components of their proper moments in the chosen direction should have the given values, which are $+\frac{1}{2}\frac{h}{2\pi}$ for ψ_α and $-\frac{1}{2}\frac{h}{2\pi}$ for ψ_β. Methods

$$\left.\begin{aligned}\psi^{\text{sym.}}\alpha_1\alpha_2(q_1,q_2)=\psi^{\text{sym.}}\alpha_1\alpha_2(q_2,q_1),\ \psi^{\text{sym.}}\alpha_1\beta_2(q_1,q_2)=\psi^{\text{sym.}}\beta_1\alpha_2(q_2,q_1),\\ \psi^{\text{sym.}}\beta_1\alpha_2(q_1,q_2)=\psi^{\text{sym.}}\beta_1\alpha_2(q_2,q_1),\ \psi^{\text{sym.}}\beta_1\beta_2(q_1,q_2)=\psi^{\text{sym.}}\beta_1\beta_2(q_2,q_1),\end{aligned}\right\} \tag{26'}$$

$$\left.\begin{aligned}\psi^{\text{antis.}}\alpha_1\alpha_2(q_1,q_2)=-\psi^{\text{antis.}}\alpha_1\alpha_2(q_2,q_1),\ \psi^{\text{antis.}}\alpha_1\beta_2(q_1,q_2)=-\psi^{\text{antis.}}\beta_1\alpha_2(q_2,q_1),\\ \psi^{\text{antis.}}\beta_1\alpha_2(q_1,q_2)=-\psi^{\text{antis.}}\beta_1\alpha_2(q_2,q_1),\ \psi^{\text{antis.}}\beta_1\beta_2(q_1,q_2)=-\psi^{\text{antis.}}\beta_1\beta_2(q_2,q_1).\end{aligned}\right\} \tag{27'}$$

The skew-symmetric solution is also the one that fulfills the "equivalence rule" in the general case of N electrons, and is the only one that occurs in nature [2]). It seems to me to

Here is Pauli's 1927 paper which suggested that electrons have an two component form, associated with angular momentum.

He was the one who named them "α" and "β".

The Nobel-prize winning sentence is underlined.

The rotational and other symmetry arguments are based on ideas that were discussed extensively here at Erice, by Sidney Coleman, which also helped seal the standard model of particle physics.

3. Electron coordinates *x* and spin

Pauli's 1927 paper & Dirac's 1926 paper

required for its expansion. Thus the symmetrical eigenfunctions alone or the antisymmetrical eigenfunctions alone give a complete solution of the problem. The theory at present is incapable of deciding which solution is the correct one. We are able to get complete solutions of the problem which make use of less than the total number of possible eigenfunctions at the expense of being able to represent only symmetrical functions of the two electrons by matrices.

These results may evidently be extended to any number of electrons. For r non-interacting electrons with co-ordinates $x_1, y_1, z_1 \ldots, x_r, y_r, z_r$, the symmetrical eigenfunctions are

$$\Sigma_{\alpha_1 \ldots \alpha_r} \psi_{n_1}(\alpha_1)\, \psi_{n_2}(\alpha_2) \ldots \psi_{n_r}(\alpha_r), \tag{14}$$

where $\alpha_1, \alpha_2 \ldots \alpha_r$ are any permutation of the integers $1, 2 \ldots r$, while the antisymmetrical ones may be written in the determinantal form

$$\begin{vmatrix} \psi_{n_1}(1), & \psi_{n_1}(2) & \ldots \psi_{n_1}(r) \\ \psi_{n_2}(1), & \psi_{n_2}(2) & \ldots \psi_{n_2}(r) \\ \cdot\ \cdot & \cdot\ \cdot & \cdot\ \cdot\ \cdot \\ \psi_{n_r}(1), & \psi_{n_r}(2) & \ldots \psi_{n_r}(r) \end{vmatrix}. \tag{15}$$

If there is interaction between the electrons, there will still be symmetrical and antisymmetrical eigenfunctions, although they can no longer be put in these simple forms. In any case the symmetrical ones alone or the antisymmetrical ones alone give a complete solution of the problem.

Actually Dirac had already suggested that the wavefunction for electrons had to be either symmetric or anti-symmetric (see later).

Dirac visited Erice several times.

His relativistic wave equation is in front of me.

4. Calculating the wavefunction

Schrödinger wave equation

4. Calculating the wavefunction

Schrödinger wave equation

- Normally calculate Ψ by solving the Schrödinger equation (SE)

- The SE describes a travelling wave.

  Example:

  $$\Psi(x, t) = W(x - vt)$$

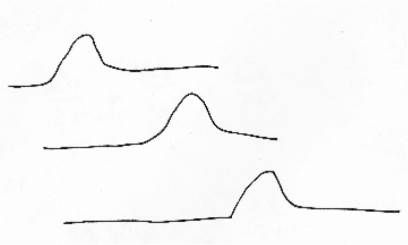
  —wave W shifts right by $d = vt$.
  —v is the velocity.

  Then, it is really easy to show:

  $$\nabla^2\Psi = \frac{1}{v^2}\frac{\partial^2\Psi}{\partial t^2}$$

  The classical wave equation

  (Use the chain rule with $u = x - vt$).



4. Calculating the wavefunction

Schrödinger wave equation

- Normally calculate Ψ by solving the Schrödinger equation (SE)

- The SE describes a travelling wave.

  Example:

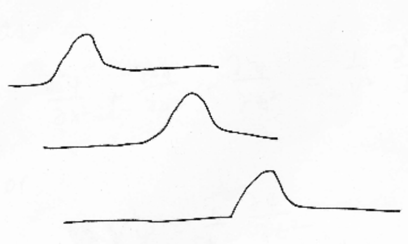
  $$\Psi(x, t) = W(x - vt)$$

  —wave W shifts right by $d = vt$.
  —v is the velocity.

  Then, it is really easy to show:

  $$\nabla^2\Psi = \frac{1}{v^2}\frac{\partial^2\Psi}{\partial t^2}$$ The classical wave equation

  (Use the chain rule with $u = x - vt$).



In the SE the velocity v each point in space changes according to the local momentum.

4. Calculating the wavefunction

Schrödinger's paper & Hamilton's optico-mechanical analogy

4. Calculating the wavefunction

Schrödinger's paper & Hamilton's optico-mechanical analogy

(q-space, not pq-space). We shall effect this generalization in a somewhat different manner in Section 7. Using the usual notations the kinetic energy T is

$$T=\tfrac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2)=(1/2m)(p_x^2+p_y^2+p_z^2). \qquad (1)$$

The well-known Hamiltonian function of action W,

$$W=\int_{t_0}^{t}(T-V)dt, \qquad (2)$$

This is the principle of least action

taken as a function of the upper limit t and of the final values of the coordinates x, y, z satisfies the Hamiltonian partial differential equation,

$$\partial W/\partial t+(1/2m)[(\partial W/\partial x)^2+(\partial W/\partial y)^2+(\partial W/\partial z)/^2]+V(x,y,z,)=0. \qquad (3)$$

To solve this equation, we put as usual

$$W=-Et+S(x,y,z), \qquad (4)$$

This is the beginning of the "separation of variables" method and S is the "reduced action"

E being an integration constant, viz., the total energy, and S a function of x, y, z only. Eq. (3) may then be written

$$|\operatorname{grad} W|=[2m(E-V)]^{\frac{1}{2}}. \qquad (5)$$

Here Schrodinger notes that W forms "wavefronts" which mover with velocity u, which was also noticed earlier by Hamilton

In this form it lends itself to a very simple geometrical interpretation.

Now let the time vary, Eq. (4) shows that the system of surfaces will not vary, but that the values of the constants will travel along the normals from surface to surface with a certain speed u, given by

$$u=E/[2m(E-V)]^{\frac{1}{2}}. \qquad (6)$$

The velocity u is a function of the energy-constant E and besides, since it contains $V(x,y,z)$ is a function of the coordinates.

3. Let us return to the system of W-surfaces, dealt with in Section 1 and let us associate with them the idea of stationary sinusoidal waves whose phase is given by the quantity W, Eq. (4). The wave-function, say ψ, will be of the form

$$\begin{aligned}\psi&=A(x,y,z)\sin(W/K)\\&=A(x,y,z)\sin[-Et/K+S(x,y,z)/K],\end{aligned} \qquad (9)$$

A being an "amplitude" function. The constant K must be introduced and must have the physical dimension of *action* (energy$\times$time), since the argument of a sine must always be a pure number. Now, since the fre-

As stated above, the wave-phenomena must in this case be studied in detail. This can only be done by using an "equation of wave propagation." Which one is this to be? In the case of a single material point, moving in an external field of force, the simplest way is to try to use the ordinary wave-equation

$$\Delta\psi-\ddot{\psi}/u^2=0 \qquad (15)$$

and to insert for u the quantity given by Eq. (6), which depends on the space coordinates (through the potential energy V) and on the frequency E/h. The latter dependence restricts the use of (15) to such functions ψ as depend on the time only through the factor $e^{\pm 2\pi i t E/h}$. (A similar restriction is always imposed on the wave equation, as soon as we have dispersion.) So we shall have

$$\ddot{\psi}=-4\pi^2E^2\psi/h^2$$

Inserting this and Eq. (6) in Eq. (15) we get

$$\Delta\psi+8\pi^2m(E-V)\psi/h^2=0, \qquad (16)$$

where ψ may be assumed to depend on x, y, z only. (We omit changing the notation of the dependent variable, which we really ought to do.)

4. Calculating the wavefunction

Schrödinger's paper & Hamilton's optico-mechanical analogy

$$E = h\,f$$ Planck eqn

$$p = \frac{h}{\lambda}$$ de Broglie eqn

$$\Longrightarrow \ \frac{E}{p} = f\,\lambda \equiv v$$ Schrödinger eqn (6)

4. Calculating the wavefunction

Schrödinger's paper & Hamilton's optico-mechanical analogy

4. Calculating the wavefunction

Schrödinger’s paper & Hamilton’s optico-mechanical analogy

(q-space, not pq-space). We shall effect this generalization in a somewhat different manner in Section 7. Using the usual notations the kinetic energy T is

$$T=\tfrac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2)=(1/2m)(p_x^2+p_y^2+p_z^2). \qquad (1)$$

The well-known Hamiltonian function of action W,

$$W=\int_{t_0}^{t}(T-V)dt, \qquad (2)$$

This is the principle of least action

taken as a function of the upper limit t and of the final values of the coordinates x, y, z satisfies the Hamiltonian partial differential equation,

$$\partial W/\partial t+(1/2m)\left[(\partial W/\partial x)^2+(\partial W/\partial y)^2+(\partial W/\partial z)/^2\right]+V(x,y,z,)=0. \qquad (3)$$

To solve this equation, we put as usual

$$W=-Et+S(x,y,z), \qquad (4)$$

This is the beginning of the "separation of variables" method and S is the "reduced action"

E being an integration constant, viz., the total energy, and S a function of x, y, z only. Eq. (3) may then be written

$$|\operatorname{grad} W| = [2m(E-V)]^{\frac{1}{2}}. \qquad (5)$$

Here Schrodinger notes that W forms "wavefronts" which mover with velocity u, which was also noticed earlier by Hamilton

In this form it lends itself to a very simple geometrical interpretation.

Now let the time vary, Eq. (4) shows that the system of surfaces will not vary, but that the values of the constants will travel along the normals from surface to surface with a certain speed u, given by

$$u=E/[2m(E-V)]^{\frac{1}{2}}. \qquad (6)$$

The velocity u is a function of the energy-constant E and besides, since it contains $V(x,y,z)$ is a function of the coordinates.

3. Let us return to the system of W-surfaces, dealt with in Section 1 and let us associate with them the idea of stationary sinusoidal waves whose phase is given by the quantity W, Eq. (4). The wave-function, say ψ, will be of the form

$$\begin{aligned}\psi&=A(x,y,z)\sin(W/K)\\&=A(x,y,z)\sin[-Et/K+S(x,y,z)/K],\end{aligned} \qquad (9)$$

A being an "amplitude" function. The constant K must be introduced and must have the physical dimension of *action* (energy×time), since the argument of a sine must always be a pure number. Now, since the fre-

As stated above, the wave-phenomena must in this case be studied in detail. This can only be done by using an "equation of wave propagation." Which one is this to be? In the case of a single material point, moving in an external field of force, the simplest way is to try to use the ordinary wave-equation

$$\Delta\psi-\ddot{\psi}/u^2=0 \qquad (15)$$

and to insert for u the quantity given by Eq. (6), which depends on the space coordinates (through the potential energy V) and on the frequency E/h. The latter dependence restricts the use of (15) to such functions ψ as depend on the time only through the factor $e^{\pm 2\pi i t E/h}$. (A similar restriction is always imposed on the wave equation, as soon as we have dispersion.) So we shall have

$$\ddot{\psi}=-4\pi^2E^2\psi/h^2$$

Inserting this and Eq. (6) in Eq. (15) we get

$$\Delta\psi+8\pi^2m(E-V)\psi/h^2=0, \qquad (16)$$

where ψ may be assumed to depend on x, y, z only. (We omit changing the notation of the dependent variable, which we really ought to do.)

4. Calculating the wavefunction

Time-independent Schrödinger equation

- Normally we get Ψ by solving the Schrödinger equation

- *Separate out the time part of the wavefunction*,

  $$\Psi(\boldsymbol{x};\boldsymbol{R})T(t),$$

  to get the time-independent form of the Schrödinger equation.

- There are many wavefunction solutions, each corresponding to a different energy E.

  The energies are both discrete and continuous.

  NOTE: wavefunction anti-symmetry and boundary conditions must be imposed on top of the SE since it isn't correct, by itself.

Time-independent Schrödinger equation:

$$H\Psi = E\Psi$$

$$H = \frac{1}{2}\sum_{i=1}^{N_\mathrm{e}} \nabla_i^2$$

Kinetic energy

$$-\sum_{i=1}^{N_\mathrm{e}}\sum_{A=1}^{N_\mathrm{N}} \frac{Z_A}{|\boldsymbol{r}_i - \boldsymbol{R}_A|}$$

Nuclear attraction energy

$$+\sum_{i=1}^{N_\mathrm{e}}\sum_{j<i}^{N_\mathrm{e}} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}$$

electron repulsion energy

4. Calculating the wavefunction

Variational principle

It is impossible to calculate exact wavefunctions for systems with more than three nuclei.

4. Calculating the wavefunction

Variational principle

It is impossible to calculate exact wavefunctions for systems with more than three nuclei.

One way to progress is to use the very powerful *variational principle*.

The energy of any approximate wavefunction Φ always lies above the energy of the exact wavefunction Ψ,

$$E_{\mathrm{QM}}[\Phi] = \int \Phi^* \, H \, \Phi \, d\boldsymbol{x} \equiv \langle \Phi \, | H | \, \Phi \rangle \geq E_{\mathrm{exact}}$$

Moreover, the wavefunction Φ which best approximates the exact wavefunction Ψ is the one which minimises the energy expression $E_{\mathrm{QM}}[\Phi]$.

5. The simplest acceptable wavefunction

Hartree-Fock (or determinant) wavefunction

5. The simplest acceptable wavefunction

Hartree-Fock (or determinant) wavefunction

- The Hartree-Fock wavefunction is an antisymmetrized product of one electron spinorbitals,

$$\Psi(\boldsymbol{x}) = \mathcal{A}[\phi_1(\boldsymbol{x}_1)\ldots\phi_{N_e}(\boldsymbol{x}_{N_e})]$$

- Spinorbitals ϕ_i to be determined. Usually fitted with gaussians.

- $\mathcal{A}$ sums all permutations of the coordinates $\boldsymbol{x}_1$ with factor ± 1, according to if the permutation is even (+1) or odd (-1).

There is also a normalization factor of $1/\sqrt{N_\mathrm{e}!}$ since there are $N_\mathrm{e}!$ terms in this wavefunction. What a nightmare.

V. Fock

Example: the three electron determinant:

$$\begin{aligned}(\sqrt{3!})\Psi = {} & \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_2)\phi_3(\boldsymbol{x}_3) - \phi_1(\boldsymbol{x}_1)\phi_2(\boldsymbol{x}_3)\phi_3(\boldsymbol{x}_2) \\ & - \phi_1(\boldsymbol{x}_2)\phi_2(\boldsymbol{x}_1)\phi_3(\boldsymbol{x}_3) + \phi_1(\boldsymbol{x}_2)\phi_2(\boldsymbol{x}_3)\phi_3(\boldsymbol{x}_1) \\ & - \phi_1(\boldsymbol{x}_3)\phi_2(\boldsymbol{x}_2)\phi_3(\boldsymbol{x}_1) + \phi_1(\boldsymbol{x}_3)\phi_2(\boldsymbol{x}_1)\phi_3(\boldsymbol{x}_2)\end{aligned}$$

— When the orbitals are determined by minimising energy E_QM, we get the Hartree-Fock wavefunction.

— Can also determine orbitals using density functional theory, minimising energy E_DFT.

5. The simplest acceptable wavefunction

Practical procedure for finding the orbitals

The idea is to fit the orbitals with fitting functions G, usually gaussians, and to determine the fitting coefficients $\boldsymbol{c}$.

$$\varphi_i(\boldsymbol{r}) = \sum_{\mu=1}^{N_{\mathrm{bf}}} G_\mu(\boldsymbol{r}) c_{\mu i}$$

G $=$ Basis functions G
$\boldsymbol{c}$ = MO coefficients

$$\downarrow$$

$$\phi_i^\sigma(\boldsymbol{r}) = \varphi_i(\boldsymbol{r})\sigma(\xi)$$

Spin orbitals
$\sigma = \alpha$ or β

$$\downarrow$$

$$\Phi = \mathcal{A}[\phi_1^\alpha(\boldsymbol{x}_1)\phi_1^\beta(\boldsymbol{x}_1)\ldots]$$

Determinant wavefunctions

$$\downarrow$$

The variational minimisation requires calculating many integrals of Hamiltonian H involving the gaussian fitting functions.

$$E_{\mathrm{QM}}[\Phi[\boldsymbol{c}]] = \int \Phi\, H\, \Phi$$

QM energy function
Many gaussian integrals!

$$\downarrow$$

Minor point: the orbitals must be normalised, which leads to the appearance of orbital energy eigenvalues ***epsilon***.

$$\boldsymbol{Fc} = \boldsymbol{Sc}\epsilon$$

Minimise E_{QM} to get $\boldsymbol{c}$
$\boldsymbol{F}$ $=$ Fock mx
ϵ $=$ eigenvalues

6. The Hohenberg-Kohn theorem

6. The Hohenberg-Kohn theorem

- Hohenberg & Kohn (1964) proved (in four lines!) that the QM energy is a unique functional of the electron density.

- Later, Kohn & Sham (1965) defined a single determinant wavefunction Ψ_{KS} which had the "exact" Born-Oppenheimer electron density.

- Using this wavefunction, they could approximate the kinetic energy very well, and make good approximate energy functionals of the density.

  This has led to "density functional theory" (DFT) a theory which produces good results for quantum mechanical energies efficiently provided your systems are fairly "normal".

We shall now show that conversely $v(\mathbf{r})$ is a unique functional of $n(\mathbf{r})$, apart from a trivial additive constant.

The proof proceeds by *reductio ad absurdum*. Assume that another potential $v'(\mathbf{r})$, with ground state Ψ' gives rise to the *same* density $n(\mathbf{r})$. Now clearly [unless $v'(\mathbf{r})-v(\mathbf{r})=\text{const}$] Ψ' cannot be equal to Ψ since they satisfy different Schrödinger equations. Hence, if we denote the Hamiltonian and ground-state energies associated with Ψ and Ψ' by H, H' and E, E', we have by the minimal property of the ground state,

$$E' = (\Psi', H'\Psi') < (\Psi, H'\Psi) = (\Psi, (H+V'-V)\Psi),$$

so that

$$E' < E + \int [v'(\mathbf{r}) - v(\mathbf{r})] n(\mathbf{r}) d\mathbf{r}. \tag{6}$$

Interchanging primed and unprimed quantities, we find in exactly the same way that

$$E < E' + \int [v(\mathbf{r}) - v'(\mathbf{r})] n(\mathbf{r}) d\mathbf{r}. \tag{7}$$

Addition of (6) and (7) leads to the inconsistency

$$E + E' < E + E'. \tag{8}$$

Thus $v(\mathbf{r})$ is (to within a constant) a unique functional of $n(\mathbf{r})$; since, in turn, $v(\mathbf{r})$ fixes H we see that the full many-particle ground state is a unique functional of $n(\mathbf{r})$.

The Hohenberg & Kohn (1964). It does not give any idea *how* to obtain the magical functional.

6. The Hohenberg-Kohn theorem

An important corrollary

Therefore, the three-dimensional ground-state electron density is all that is needed to define the exact ground-state electronic wavefunction Ψ of a system. Although not often mentioned, the HohenbergKohn theorem ensures that the ground-state wavefunction is reconstructable from experiment, since the electron density is experimentally reconstructable.

Davidson & Jayatilaka (2022)

We shall now show that conversely $v(\mathbf{r})$ is a unique functional of $n(\mathbf{r})$, apart from a trivial additive constant.

The proof proceeds by *reductio ad absurdum*. Assume that another potential $v'(\mathbf{r})$, with ground state Ψ' gives rise to the *same* density $n(\mathbf{r})$. Now clearly [unless $v'(\mathbf{r})-v(\mathbf{r})=\text{const}$] Ψ' cannot be equal to Ψ since they satisfy different Schrödinger equations. Hence, if we denote the Hamiltonian and ground-state energies associated with Ψ and Ψ' by H, H' and E, E', we have by the minimal property of the ground state,

$$E'=(\Psi',H'\Psi')<(\Psi,H'\Psi)=(\Psi,(H+V'-V)\Psi),$$

so that

$$E'<E+\int[v'(\mathbf{r})-v(\mathbf{r})]n(\mathbf{r})d\mathbf{r}. \tag{6}$$

Interchanging primed and unprimed quantities, we find in exactly the same way that

$$E<E'+\int[v(\mathbf{r})-v'(\mathbf{r})]n(\mathbf{r})d\mathbf{r}. \tag{7}$$

Addition of (6) and (7) leads to the inconsistency

$$E+E'<E+E'. \tag{8}$$

Thus $v(\mathbf{r})$ is (to within a constant) a unique functional of $n(\mathbf{r})$; since, in turn, $v(\mathbf{r})$ fixes H we see that the full many-particle ground state is a unique functional of $n(\mathbf{r})$.

The Hohenberg & Kohn (1964). It does not give any idea *how* to obtain the magical functional.

7. Experimental wavefunctions

7. Experimental wavefunctions

There are two general ways to obtain wavefunctions from experiment

1. Reconstruction methods.
   Also called tomographic methods.
2. Weak measurement methods.

Shown is a measurement (not even a reconstruction!) of the H atom 1s wavefunction.

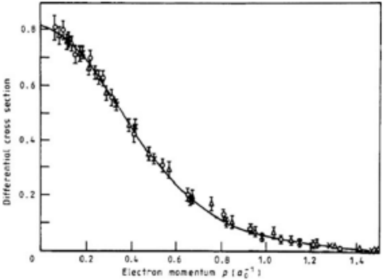


Figure 6. The theoretical and (e,2e) experimentally determined momentum-space wavefunction for the H atom. From Lohman & Weigold (1981). See also McCarthy and Weigold (1988)

7. Experimental wavefunctions

There are two general ways to obtain wavefunctions from experiment

1. Reconstruction methods.
   Also called tomographic methods.
2. Weak measurement methods.

IMPORTANT

The reconstruction methods we use involve theoretical assumptions and models which, although tested and benchmarked, may not be entirely relevant to your system.

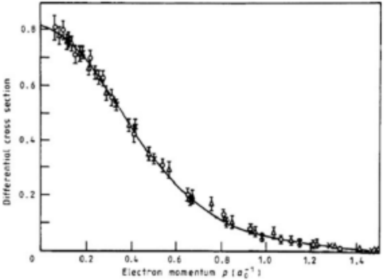


Figure 6. The theoretical and (e,2e) experimentally determined momentum-space wavefunction for the H atom. From Lohman & Weigold (1981). See also McCarthy and Weigold (1988)

7. Experimental wavefunctions

There are two general ways to obtain wavefunctions from experiment

1. Reconstruction methods.
   Also called tomographic methods.
2. Weak measurement methods.

IMPORTANT

The reconstruction methods we use involve theoretical assumptions and models which, although tested and benchmarked, may not be entirely relevant to your system.

Especially, the most important science involves unusual systems, investigating outliers, misfits, and weird stuff. Where we can learn, or exploit.

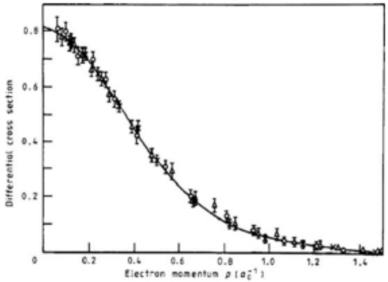


Figure 6. The theoretical and (e,2e) experimentally determined momentum-space wavefunction for the H atom. From Lohman & Weigold (1981). See also McCarthy and Weigold (1988)

7. Experimental wavefunctions

The X-ray (or ED) structure factors contain detailed information on the electron density which can be used to reconstruct a reasonable wavefunction.

But how do we do it?

7. Experimental wavefunctions

How to do it

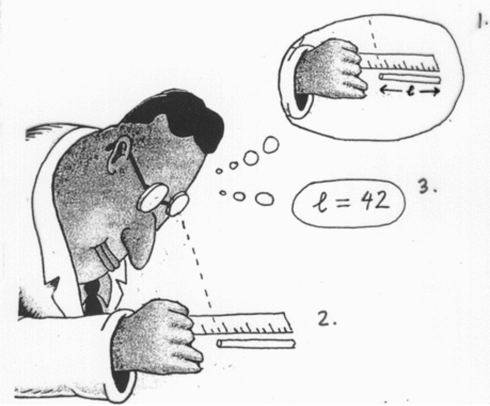
Do it like any measurement.

❶ Think of a model with parameters.

Think of an experiment to find those parameters.

❷ Do the experiment

❸ Find the parameters



7. Experimental wavefunctions

How to do it

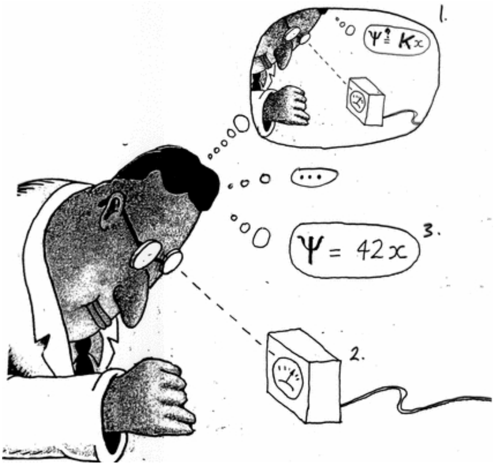
Do it like any measurement.

1. Think of a model with parameters.

   Think of an experiment to find those parameters.

2. Do the experiment

3. Find the parameters



7. Experimental wavefunctions

Should we do this?

7. Experimental wavefunctions

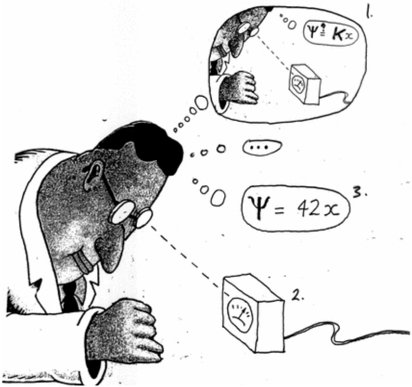
Should we do this?

We want to model experiments well.

The wavefunction is the best model: it holds all the information in a system.

It can predict the results of multiple experiments *simultaneously*.

The wavefunction is perhaps the ultimate framework in which to develop any model, theoretical or experimental.



7. Experimental wavefunctions

Why are we doing this?

7. Experimental wavefunctions

Why are we doing this?

We are doing this to find new things.

To find new things you sometimes need better accuracy.

There are two cases of interest:

1. You have crap data, you are desperate, and you want to see if you can improve it.

   Example: protein X-ray crystallography

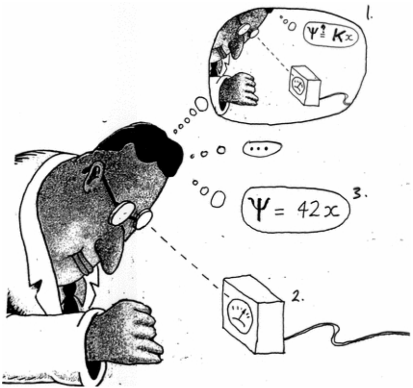
   Note: First try getting better data

2. You have good data and you don't understand why other models aren't working.

   So try wavefunction fitting.

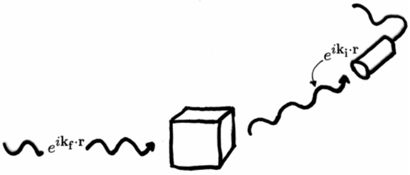
   (a) If it agrees—problem solved.

   (b) No agreement?—hooray!—Nobel prize



7. Experimental wavefunctions

The experiment and the model



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector

X-ray diffraction measurement.

Initial wavefunction:

$$e^{i\boldsymbol{k}_i\cdot\boldsymbol{r}}\Psi_{\text{crystal}}$$

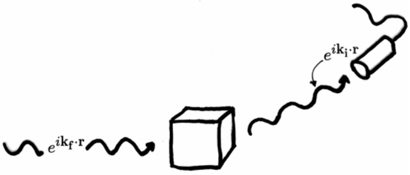
Final wavefunction after measurement:

$$e^{i\boldsymbol{k}_f\cdot\boldsymbol{r}}\Psi_{\text{crystal}}$$

Elastic scattering: crystal does not change.

7. Experimental wavefunctions

The experiment and the model



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector

X-ray diffraction measurement.

Initial wavefunction:

$$e^{i\boldsymbol{k}_i\cdot\boldsymbol{r}}\Psi_{\text{crystal}}$$

Final wavefunction after measurement:

$$e^{i\boldsymbol{k}_f\cdot\boldsymbol{r}}\Psi_{\text{crystal}}$$

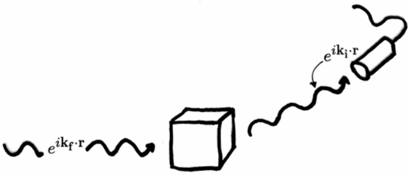
Elastic scattering: crystal does not change.

Only the direction of the X-ray plane-wave vector changes, from $\boldsymbol{k}_i$ to $\boldsymbol{k}_f$.

(Yes, the picture is wrong way round).

7. Experimental wavefunctions

The experiment and the model



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector. Note that because the crystal is periodic only certain scattering directions are allowed. These are seen as diffraction spots, and their direction and intensity is measured.

Example (cont'd).

Probability for this experiment event is:

$$
\begin{aligned}
&|\langle e^{i\boldsymbol{k}_i\cdot\boldsymbol{r}}\Psi_{\text{crystal}}|e^{i\boldsymbol{k}_f\cdot\boldsymbol{r}}\Psi_{\text{crystal}}\rangle|^2 \\
&= \int e^{i(\boldsymbol{k}_f-\boldsymbol{k}_i)\cdot\boldsymbol{r}}|\Psi_{\text{crystal}}|^2 \\
&\approx \int e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\rho(\boldsymbol{r}) \\
&\equiv F(\boldsymbol{k}) \qquad \text{structure factor}
\end{aligned}
$$

and where

$$\boldsymbol{k} = \boldsymbol{k}_f - \boldsymbol{k}_i \qquad \text{scattering vector}$$

$$\rho(\boldsymbol{r}) \approx |\Psi_{\text{crystal}}|^2(\boldsymbol{r}) \qquad \text{electron density} \quad .$$

7. Experimental wavefunctions

Formula for the structure factor from a determinant wavefunction Φ

$$F_k^{\text{calc}}[\Phi] = s|F^{\text{model}}(\boldsymbol{q}_k)|$$

$$F^{\text{model}}(\boldsymbol{q}_k) = \sum_A^{\text{atoms}} n_A^{-1} \sum_{\{\boldsymbol{S},\boldsymbol{t}\}}^{\text{symops}} \bar{f}_A(\boldsymbol{S}^T\boldsymbol{q}_k)e^{i\boldsymbol{q}_k\cdot\boldsymbol{R}_A}e^{i\boldsymbol{q}_k\cdot\boldsymbol{t}}$$

where:

$$\begin{aligned}
s &= \text{Scale factor} \\
\boldsymbol{q}_k &= k\text{-th scattering vector} = 2\pi\boldsymbol{A}\boldsymbol{h} \\
\boldsymbol{R}_A &= \text{Asymmetric atom coordinate} \\
\{\boldsymbol{S},\boldsymbol{t}\} &= \text{Roto-translational symmetry operators} \\
n_A &= \text{Atom site symmetry factor} \\
\bar{f}_A(\boldsymbol{q}_k) &= \text{Thermally-averaged aspherical form factor} \\
&= f_A(\boldsymbol{q}_k;\{\boldsymbol{R}_A\};\Phi)\times\text{TF}(\boldsymbol{q}_k) \\
f_A(\boldsymbol{q}_k;\{\boldsymbol{R}_A\};\Phi) &= \text{Static aspherical form factor; from } \Phi \\
\text{TF}(\boldsymbol{k}) &= \text{Temperature factor (Gram-Charlier form)} \\
&= e^{-\frac{1}{2!}\boldsymbol{U}^2:\boldsymbol{k}}\left[1-\frac{i}{3!}\boldsymbol{U}^3 \vdots \boldsymbol{k}+\frac{1}{4!}\boldsymbol{U}^4 :: \boldsymbol{k}\right] \\
\boldsymbol{U}^2,\boldsymbol{U}^3,\boldsymbol{U}^4 &= \text{Atomic displacement parameters, ADPs}
\end{aligned}$$

7. Experimental wavefunctions

Idea: how we fit Φ with not enough data

7. Experimental wavefunctions

Idea: how we fit Φ with not enough data

- We do a *constrained fit*.

  In crystallography, this is also called a *restrained fit*.

- We minimize the Lagrangian

$$L = E_{QM}[\boldsymbol{c}] + \lambda \mathrm{GoF}^2[\boldsymbol{c}]$$

  with respect to the orbital coefficients $\boldsymbol{c}$.

- The term $\lambda\,\mathrm{GoF}^2$ is the least-squares penalty;
  it enforces the fitting constraint.

7. Experimental wavefunctions

Idea: how we fit Φ with not enough data

- We do a *constrained fit*.

  In crystallography, this is also called a *restrained fit*.

- We minimize the Lagrangian

$$L = E_{QM}[\boldsymbol{c}] + \lambda \mathrm{GoF}^2[\boldsymbol{c}]$$

  with respect to the orbital coefficients $\boldsymbol{c}$.

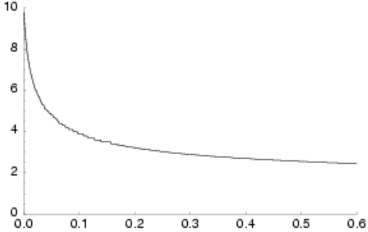
- The term $\lambda\,\mathrm{GoF}^2$ is the least-squares penalty;
  it enforces the fitting constraint.

Note:

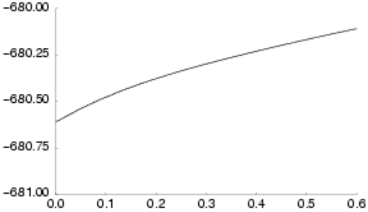
- Choosing $\lambda = 0$ gives a normal QM calculation.
- Choosing $\lambda = \infty$ gives a “regularized” least squares fit.
- In principle we can get a perfect fit!
- In principle we need no experimental data.
  (But then, we do not have an experimental model!).

7. Experimental wavefunctions

X-ray constrained wavefunctions (XCW)



GoF^2 vs λ



E vs λ

7. Experimental wavefunctions

How to choose λ and end the fit

You are finished if:

- χ^2 is as small as you can get
- And: χ^s_{free} has not increased
- And: you get *reasonable fit indicators* (see later)
- And: you didn't notice anything else suspicious.
- *Don't report λ, it is meaningless*. Report the χ^2.

The notes discuss in more detail certain issues concerning this difficult problem, which I have glossed over!

The oxalic acid experimental wavefunction

From Grimwood & Jayatilaka (2001)

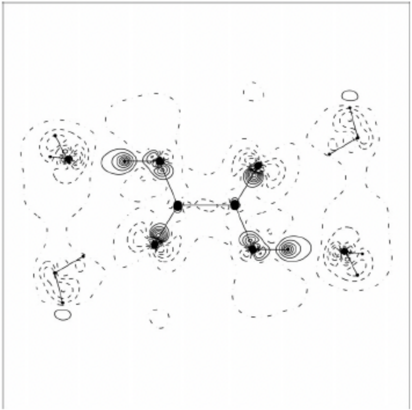


Figure 12
Difference between the constrained ($\lambda = 0.3$) and unconstrained Hartree–Fock static density plots for an oxalic acid unit surrounded by four water molecules.

Effect on electron density of doing an X-ray constrained Hartree-Fock (XCHF) calculation.

Contours at 0.1 e Å^{-3}.

The oxalic acid experimental wavefunction

From Grimwood & Jayatilaka (2001)

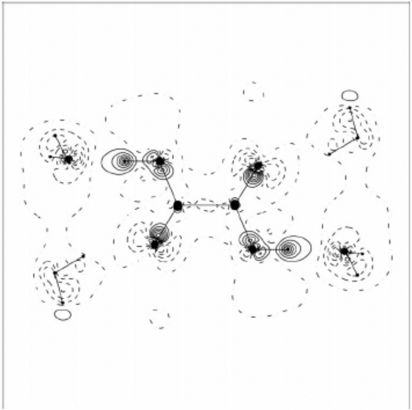


Figure 12
Difference between the constrained ($\lambda = 0.3$) and unconstrained Hartree–Fock static density plots for an oxalic acid unit surrounded by four water molecules.

Effect on electron density of doing an X-ray constrained Hartree-Fock (XCHF) calculation.

Contours at 0.1 e Å^{-3}.

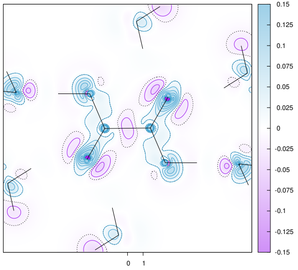
After fitting, there are strangely enhanced hydrogen bonds in this picture!

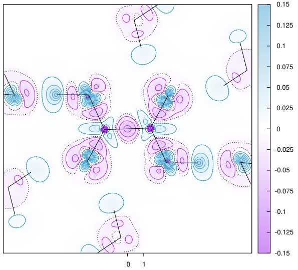
It suggests that the geometrical parameters for the hydrogen atoms were not quite right!

Some results

The oxalic acid experimental wavefunction

From Davidson, Grabowsky & Jayatilaka (2022)





Effect on ED do to fitting HF wavefunction simultaneously to 12 data different sets.

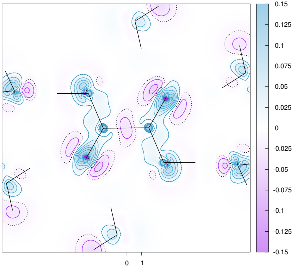
Contours at 0.025 e Å^{-3}.

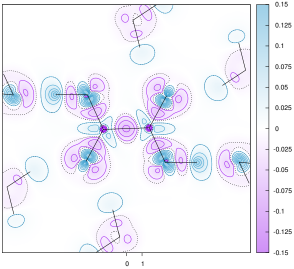
Top: XCHF wavefunction

Bottom: Calculation: CCSD-HF with pt. charges

The oxalic acid experimental wavefunction

From Davidson, Grabowsky & Jayatilaka (2022)





Effect on ED do to fitting HF wavefunction simultaneously to 12 data different sets.

Contours at 0.025 e Å^{-3}.

Top: XCHF wavefunction

Bottom: Calculation: CCSD-HF with pt. charges

Good qualitative agreement of electron density changes ’ due to crystal and correlated effects.

–Except near the atom centers.

– CC bond e- depletion is exaggerated in theory.

– H atom e- buildup persists in theory but is now much smaller.

Molecular polarizabilities and refractive indices

Jayatilaka, Munshi, Turner, Howard, Spackman (2009)

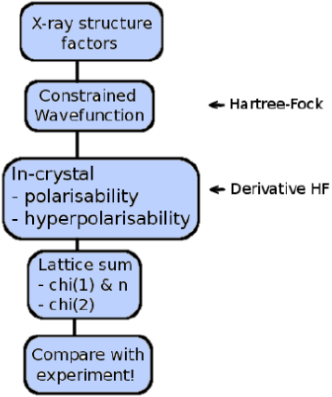


Fig. 11 Outline scheme for calculating refractive indices and NLO properties from X-ray data. See ref [15] for details.

Molecular polarizabilities and refractive indices

Jayatilaka, Munshi, Turner, Howard, Spackman (2009)

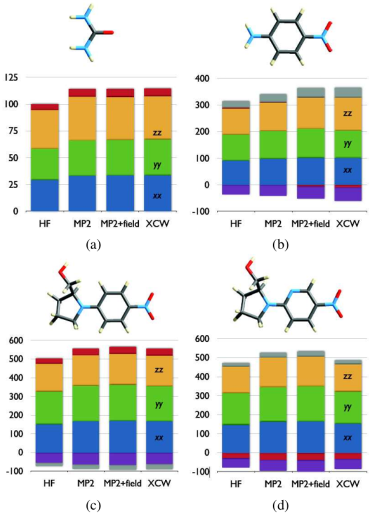


Fig. 12 Components of the in-crystal molecular polarisabily tensor (/au) for selected non-linear molecules (a) Urea (b) pNA (c) NPP and (d) PNP for increasingly accurate levels of theory (HF: Hartree-Fock, MP2: second order Møller-Plesset theory, MP2 + field: an MP2 calculation in a static electric field calculated by Ewald summation of surrounding molecular dipole moments) compared with the X-ray constrained Hartree-Fock wavefunction (XCW). For details of axis system see **??**

Molecular polarizabilities and refractive indices

Jayatilaka, Munshi, Turner, Howard, Spackman (2009)

	axis	HF	MP2	MP2 +field	XCW	Expt
urea	$n_a = n_b$	1.39	1.44	1.45	1.45	1.477
	n_c	1.52	1.62	1.59	1.60	1.583
POM	n_a	1.55	1.58	1.58	1.55	1.625
	n_c	1.58	1.62	1.62	1.60	1.663
	n_b	1.71	1.82	1.81	1.78	1.829
NPP	n_X	1.42	1.45	1.44	1.45	1.457
	$n_Y = n_b$	1.63	1.69	1.70	1.67	1.774
	n_Z	1.71	1.80	1.82	1.79	1.829
PNP	n_X	1.43	1.45	1.45	1.43	1.456
	$n_Y = n_b$	1.62	1.68	1.69	1.61	1.732
	n_Z	1.70	1.80	1.82	1.74	1.880

Refractive indices *n*.

Don't look: the results from fitted wavefunctions are mostly better when compared to experiment or better calculations.

Molecular polarizabilities and refractive indices

Jayatilaka, Munshi, Turner, Howard, Spackman (2009)

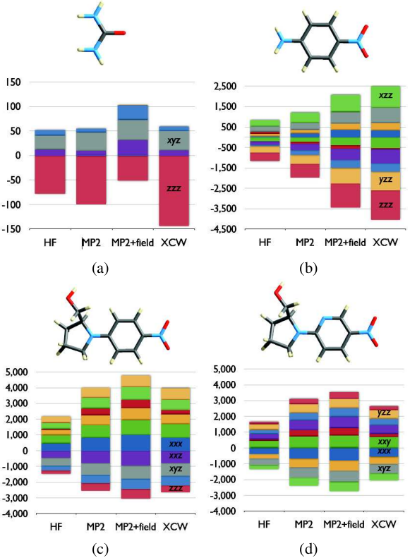


Fig. 13 Components of the in-crystal molecular hyper-polarisability tensor (/au) for selected non-linear molecules (a) Urea (b) pNA (c) NPP and (d) PNP for increasingly accurate levels of theory, as described in previous figures, compared with the X-ray constrained Hartree-Fock wavefunction (XCW). For details of axis system see **??**